

Patient Health Record Consent Management System (PHRCMS)

Software Engineering Lab 1 — Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a patient to grant time-bounded access permissions for specific diagnostic records to verified clinic doctors, requiring a valid expiration date and time.	High	A verified clinic doctor can view the specified diagnostic records within the designated consent time-frame, and the system automatically blocks access immediately after the consent's expiration timestamp.	Enables patients to share health data temporarily, minimizing data exposure risks.
FR-002	Functional	The system shall allow a patient to revoke any active consent permission at any time prior to its scheduled expiration.	High	Upon revocation by the patient, the status of the consent is immediately updated to 'revoked', and any subsequent attempt by the authorized doctor to access the record is blocked.	Gives patients real-time control over their data, aligning with patient-centric consent principles.
FR-003	Functional	The system shall display a real-time list of all active and expired consent permissions to the logged-in patient, including doctor details, record names, and expiration timestamps.	Medium	The consent registry page lists all consents, showing doctor name, record ID, status (active/expired/revoked), and remaining duration (or expiry date/time) accurately.	Promotes transparency, allowing patients to audit who has access to their data.
FR-004	Functional	The system shall permit a verified clinic doctor to view authorized patient diagnostic records only if there is an active, unexpired consent granted by the patient.	High	If a doctor attempts to access a record, the system grants access if an active consent exists, and denies access with an error if consent is missing, expired, or revoked.	Enforces the consent policy, preventing unauthorized data access.

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FR-005	Functional	The system shall allow a Clinic Administrator to register and manage the verification status of clinic doctors in the system.	Medium	The administrator can add a doctor to the registry and toggle their status as 'verified' or 'unverified'. Unverified doctors cannot be selected by patients for consent grants.	Ensures that patients only grant access to legitimate, authorized medical professionals.
NFR-001	Nonfunctional	All consent grant, revoke, and access events shall be permanently written to an append-only audit trail.	High	Every consent action and record-access event creates a log entry with timestamp, actor ID, and action type, which cannot be modified or deleted by any user or administrator.	Ensures accountability, data integrity, and compliance with medical privacy audits.
NFR-002	Nonfunctional	The system shall complete the verification of consent status and authorize or deny a doctor's access request in less than 500 milliseconds.	High	Under automated performance testing with 50 concurrent requests, the average response time to verify consent status and return the access decision is less than 500 milliseconds.	Enforces prompt access decision checks, ensuring doctor workflow is not delayed while maintaining strict security checks.